



**UNIVERSITY OF
CALGARY**

ENGG 684- Introduction to Project Management

Segment 2: Project Scheduling and Subcontractor Selection

Shaktipal Singh Deora (UCID 30239938)

Group – 8, M&M Constructions Ltd.

Course Instructor:

Estacio Pereira, PhD

Assistant Professor (Teaching) in Project Management

Department of Civil Engineering – Schulich School of Engineering

1. Detailed Schedule for 08 Projects

The Detailed Schedule for the 08 Projects are presented in the Gantt Chart. The Gantt Chart is representing the major tasks, their start and finish dates for each project. The Gantt chart also represent the selection of the subcontractors, which we will discuss further in the report providing the justification for the selection criteria of the subcontractors:

[illegible]

2. Subcontractor Selection Table

After doing the detailed analysis and applying some permutation and combination, the decision on the Subcontractor selection is finalised and table for which is attached below:

Table 1: Subcontractor Selection Table

Sr. No.	Task	Selected Subcontractor	Production rate (units/week) average	Cost/Week (\$)
1	Architectural Drawing	Arch ContB	14 pages	10,000
2	Geotechnical	Geo ContB	25 boreholes	9,000
3	Structural Drawing	Struct ContA	12 sheets	9,000
4	Electrical	Elec ContA	25 circuits	10,000
5	Mechanical	Mech ContA	5 zones	8,000
6	Plumbing	Plumb ContA	8 fixtures	7,000
7	Compatibility Check	Comp ContB	15 design packages	8,000

3. Justification For Subcontractor Choice

To get started, I first calculated how much time it would take to complete all the tasks using each contractor individually, Contractor A, Contractor B, and Contractor C.

- **Contractor C** could finish all tasks in 32 weeks
- **Contractor B** would take 37 weeks
- **Contractor A** would exceed 40 weeks, which is beyond our schedule

So, it was clear that Contractor A alone wasn't a viable option. Ideally, Contractor C was the best in terms of time, but then I also considered the direct costs, the direct cost for each Contractor is mentioned below and the direct cost calculation table is attached in **Table 2** below :

- Contractor A: \$1,350,000
- Contractor B: \$1,432,500
- Contractor C: \$1,929,000

Although Contractor C could finish early, it had the highest cost. I also calculated the indirect cost savings, which came out to be \$14,375 per week saved if the project finishes early and the table for indirect cost calculation is attached below in **Table 3** .

So, I looked for a balance between cost and time. I tried multiple simulations and the simulation which worked best for me is the following one:

- Contractor B handled: Architectural Drawing and Geotechnical
- Contractor A handled: Structural Drawing, Electrical, Mechanical, and Plumbing
- Contractor C handled: Compatibility Check

This setup completed the project in 36 weeks, but the direct cost was \$1,604,000 and indirect cost is \$517,500 and total cost will be \$2,121,500, which was still relatively high. So, I decided to remove Contractor C from the mix and reassign Compatibility Check to Contractor B.

This new setup still completed everything in 36 weeks but reduced the direct cost to \$1,355,000 and indirect cost is \$517,500 and total cost will be \$1,872,500, making it the most cost-effective and time-efficient option.

Final Contractor Assignment:

- Architectural Drawing: Contractor B
- Geotechnical: Contractor B
- Structural Drawing: Contractor A
- Electrical: Contractor A
- Mechanical: Contractor A
- Plumbing: Contractor A
- Compatibility Check: Contractor B

Table 2: Direct Cost Calculation table

Task	Suncontractor	Direct cost for Project 1	Direct cost for Project 2	Direct cost for Project 3	Direct cost for Project 4	Direct cost for Project 5	Direct cost for Project 6	Direct cost for Project 7	Direct cost for Project 8	Total Sum
Architectural Drawing	Arch ContA	24,000	32,000	40,000	24,000	32,000	40,000	24,000	48,000	264,000
	Arch ContB	20,000	30,000	40,000	30,000	30,000	40,000	20,000	50,000	260,000
	Arch ContC	26,000	39,000	52,000	26,000	39,000	39,000	26,000	52,000	299,000
Geotechnical	Geo ContA	14,000	21,000	28,000	21,000	21,000	28,000	35,000	35,000	203,000
	Geo ContB	18,000	27,000	36,000	18,000	18,000	27,000	36,000	36,000	216,000
	Geo ContC	22,000	22,000	33,000	22,000	22,000	33,000	44,000	33,000	231,000
Structural Drawing	Struct ContA	9,000	27,000	27,000	18,000	18,000	18,000	18,000	27,000	162,000
	Struct ContB	11,000	22,000	22,000	11,000	22,000	22,000	22,000	33,000	165,000
	Struct ContC	14,000	28,000	28,000	14,000	14,000	28,000	28,000	28,000	182,000
Electrical	Elec ContA	10,000	30,000	30,000	20,000	20,000	30,000	10,000	30,000	180,000
	Elec ContB	13,000	39,000	39,000	26,000	26,000	39,000	13,000	39,000	234,000
	Elec ContC	17,000	34,000	51,000	34,000	34,000	34,000	17,000	51,000	272,000
Mechanical	Mech ContA	16,000	24,000	32,000	24,000	24,000	32,000	16,000	32,000	200,000
	Mech ContB	20,000	30,000	30,000	20,000	30,000	30,000	20,000	40,000	220,000
	Mech ContC	15,000	30,000	30,000	30,000	30,000	30,000	30,000	45,000	240,000
Plumbing	Plumb ContA	14,000	21,000	28,000	14,000	21,000	21,000	14,000	28,000	161,000
	Plumb ContB	9,500	19,000	28,500	19,000	19,000	19,000	19,000	28,500	161,500
	Plumb ContC	20,000	40,000	40,000	40,000	40,000	40,000	20,000	40,000	280,000
Compatibility Check	Comp ContA	12,000	24,000	30,000	18,000	18,000	24,000	18,000	36,000	180,000
	Comp ContB	16,000	24,000	32,000	16,000	16,000	24,000	16,000	32,000	176,000
	Comp ContC	25,000	50,000	75,000	50,000	50,000	50,000	50,000	75,000	425,000
Total Cost for Contractor A										1,350,000
Total Cost for Contractor B										1,432,500
Total Cost for Contractor C										1,929,000

Table 3: Indirect Cost Calculation Table

Category	Description	Quantity	Unit Cost per Week (\$)	Subtotal Week
Office Rent	Lease, utilities, shared workspace	2 rooms	5000	10000
Software Licenses	BIM, CAD, scheduling, simulation tools	8 licenses	100	800
Administrative Staff Salaries	Project coordinator, HR, finance	2 staff	800	1600
Office Supplies & Printing	Stationery, printing, largeformat plotting	Flat weekly cost	500	500
IT Infrastructure & Support	Cloud storage, hardware, technical support	3 systems	60	180
Communication Tools	PM software, video conferencing tools	3 licenses	30	90
Professional Liability Insurance	Design liability coverage	1 company policy	135	135
Training & Professional Development	Weekly average for webinars, training, certifications	Flat weekly cost	70	70
Miscellaneous Overhead	Travel, subscriptions, small equipment, snacks	Flat weekly allowance	1000	1000
Total Cost per week				14375

4. Production rate calculations used to determine task durations

The very first step to complete all the above calculations starts with production rate calculations, which we further used to determine the task durations and I also made sure that we round off up to the upper limit for the calculated weeks, so first I made a table of the data provided which I will use for the calculations to calculate production rate which is attached below in **Table 4**

Table 4: Data Provided to calculate production rate and calculate weeks.

Task	Subcontractor	Production rate (units/week) average	Cost/Week (\$)	Project 1 — Residential Home	Project 2 Office Building	Project 3 — Apartment Complex	Project 4- Health Clinic	Project 5 — Retail Strip Mall	Project 6 — Elementary School	Project 7 — Underground Parkade
Architectural Drawing	Arch ContA	10	8,000	25	40	50	30	35	45	28
	Arch ContB	14	10,000	25	40	50	30	35	45	28
	Arch ContC	16	13,000	25	40	50	30	35	45	28
Geotechnical	Geo ContA	20	7,000	40	60	80	50	50	70	100
	Geo ContB	25	9,000	40	60	80	50	50	70	100
	Geo ContC	30	11,000	40	60	80	50	50	70	100
Structural Drawing	Struct ContA	12	9,000	12	25	30	15	18	22	20
	Struct ContB	15	11,000	12	25	30	15	18	22	20
	Struct ContC	18	14,000	12	25	30	15	18	22	20
Electrical	Elec ContA	25	10,000	20	60	70	35	40	55	25
	Elec ContB	27	13,000	20	60	70	35	40	55	25
	Elec ContC	33	17,000	20	60	70	35	40	55	25
Mechanical	Mech ContA	5	8,000	8	15	18	12	14	17	10
	Mech ContB	6	10,000	8	15	18	12	14	17	10
	Mech ContC	9	15,000	8	15	18	12	14	17	10
Plumbing	Plumb ContA	8	7,000	10	22	26	16	20	24	14
	Plumb ContB	12	9,500	10	22	26	16	20	24	14
	Plumb ContC	15	20,000	10	22	26	16	20	24	14
Compatibility Check	Comp ContA	10	6,000	20	40	50	30	30	40	30
	Comp ContB	15	8,000	20	40	50	30	30	40	30
	Comp ContC	20	25,000	20	40	50	30	30	40	30

And after that we calculated the total weeks of time/ Production rate for each contractor to perform each task and is attached below in **Table 5**

Table 5: Calculated weeks for each contractor

Task	Subcontractor	Weeks Rounded for Project 1	Weeks Rounded for Project 2	Weeks Rounded for Project 3	Weeks Rounded for Project 4	Weeks Rounded for Project 5	Weeks Rounded for Project 6	Weeks Rounded for Project 7	Weeks Rounded for Project 8
Architectural Drawing	Arch ContA	3	4	5	3	4	5	3	6
	Arch ContB	2	3	4	3	3	4	2	5
	Arch ContC	2	3	4	2	3	3	2	4
Geotechnical	Geo ContA	2	3	4	3	3	4	5	5
	Geo ContB	2	3	4	2	2	3	4	4
	Geo ContC	2	2	3	2	2	3	4	3
Structural Drawing	Struct ContA	1	3	3	2	2	2	2	3
	Struct ContB	1	2	2	1	2	2	2	3
	Struct ContC	1	2	2	1	1	2	2	2
Electrical	Elec ContA	1	3	3	2	2	3	1	3
	Elec ContB	1	3	3	2	2	3	1	3
	Elec ContC	1	2	3	2	2	2	1	3
Mechanical	Mech ContA	2	3	4	3	3	4	2	4
	Mech ContB	2	3	3	2	3	3	2	4
	Mech ContC	1	2	2	2	2	2	2	3
Plumbing	Plumb ContA	2	3	4	2	3	3	2	4
	Plumb ContB	1	2	3	2	2	2	2	3
	Plumb ContC	1	2	2	2	2	2	1	2
Compatibility Check	Comp ContA	2	4	5	3	3	4	3	6
	Comp ContB	2	3	4	2	2	3	2	4
	Comp ContC	1	2	3	2	2	2	2	3